

Principles of Economics

Lecture 7: Markets, Efficiency and Public Policy

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Plan for today's class

Introduction

Externalities: Definition

Externalities: Analysis

Externalities: Solutions

Externalities and Climate Change

Asymmetric Information

Public Goods

Limits to Markets

Summary

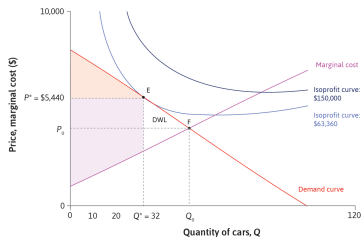
1. Introduction

- ▶ Different market conditions, different outcomes:
 - ▶ Week 5: price-setting firms, an equilibrium that is not Pareto efficient (a market failure);
 - ▶ Week 6: price-taking firms, a Pareto efficient equilibrium.
- ▶ Other market failures:
 - ▶ Misuse of insecticides in agriculture.
 - ▶ Misuse of antibiotics.
- ▶ We want to know:
 - ▶ Why do inefficiencies occur?
 - ▶ To what extent can the market achieve the best allocation of resources?
 - ▶ How can the government intervene to correct market failures?
- ▶ **Market failure:** the market does not reach the best possible allocation.

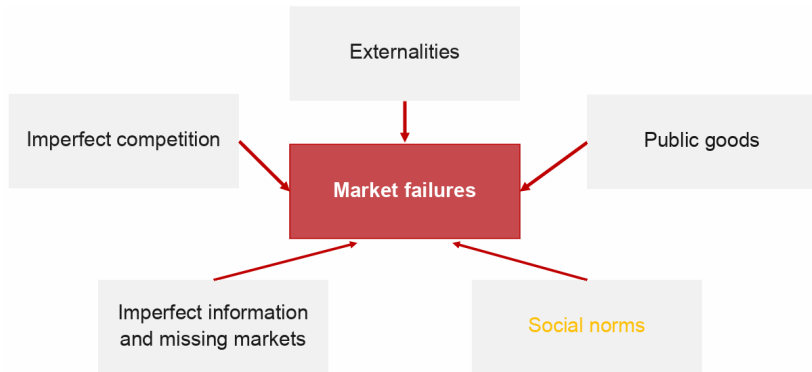
- ▶ A firm with power to set the price puts it above marginal cost:

- ▶ **Market power:** one or a few firms share the market:

- ▶ There are barriers to entry.
- ▶ There are economies of scale (*natural monopoly*);
- ▶ The product is differentiated;



- ▶ Competition (antitrust) policy attacks the **deadweight loss**.



- ▶ When the market fails, whoever decides often does not bear all the consequences;
- ▶ Others bear the costs of those decisions.

2. Externalities: Definition

Definition

- ▶ An **externality**, positive or negative, occurs when the decisions of a person or a firm affect others in their **production or consumption**, and no contract covers the harm or the benefit.
- ▶ **Examples:**
 - ▶ Which ones are positive externalities and which are negative?
 1. Car pollution
 2. Technological innovations
 3. Tobacco
 4. Urban construction
 5. Household waste
 6. Honey
 7. Vaccines
 8. Traffic jams
 9. Education

Definition

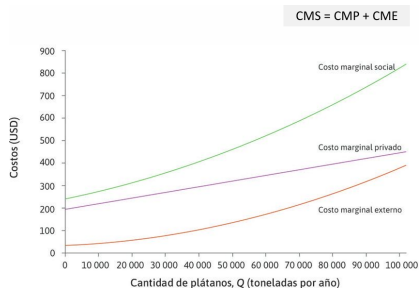
- ▶ An **externality**, positive or negative, occurs when the decisions of a person or a firm affect others in their **production or consumption**, and no contract covers the harm or the benefit.
- ▶ **Examples:**
 - ▶ Which are externalities in production and which in consumption?
 1. Car pollution
 2. Technological innovations
 3. Tobacco
 4. Urban construction
 5. Household waste
 6. Honey
 7. Vaccines
 8. Traffic jams
 9. Education

3. Externalities: Analysis

- ▶ **Negative externalities:** whoever causes them does not bear the cost;
- ▶ We focus on production; externalities in consumption are analogous.
- ▶ Three concepts for the analysis:
 - ▶ **Marginal private cost (MPC):** Marginal cost for the agent who causes the externality.
Example: what the insecticide costs the farmer.
 - ▶ **Marginal external cost (MEC):** Marginal cost caused by the agent who decides or acts.
Example: what the fishermen lose as their fisheries shrink.
 - ▶ **Marginal social cost (MSC):** $MSC = MPC + MEC$
- ▶ **A negative externality exists when $MSC > MPC$.**
- ▶ **A positive externality exists when $MSC < MPC$.**

- ▶ The three concepts that measure the externality:
 - ▶ Vertical axis: Marginal costs;
 - ▶ Horizontal axis: quantity of output.

- ▶ Here the marginal external cost is below the marginal private cost.
- ▶ The MEC rises: the external effects accumulate.



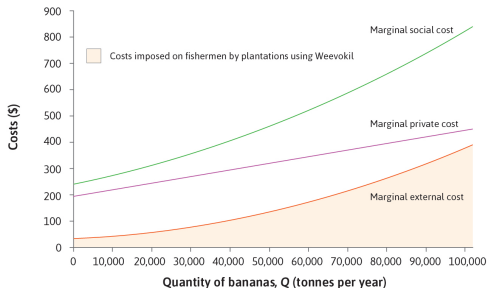
- ▶ A price-taking firm produces where price equals marginal cost:

$$P = MPC$$

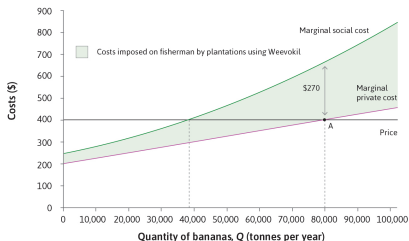
- ▶ But society's optimum is at:

$$P = MSC$$

- ▶ **Negative externalities** cause **overproduction**: too much insecticide.



- ▶ A price-taking firm produces where price equals marginal cost:
- ▶ Producing 80 thousand tonnes is not Pareto efficient:
- ▶ At 80 thousand tonnes the marginal external cost is \$270 per tonne (marginal social cost \$400 + \$270 = \$670);
- ▶ Fishermen would pay the farmers up to \$270 for that last tonne not to be produced.
- ▶ But the farmers care about MPC, not MSC.



4. Externalities: Solutions

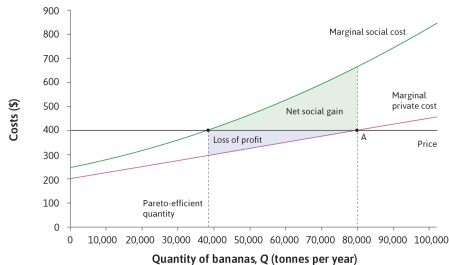
- ▶ The market fails because complete contracts cannot be **negotiated**.
- ▶ **Contracts are incomplete**: they cannot pin down every effect of the exchange in an enforceable way;
- ▶ The usual problem is missing information:
 - ▶ Either it is asymmetric (only the producer knows the level of production), or the cost of the externality is simply hard to estimate.
- ▶ Another one: **property rights are not clearly defined**:
 - ▶ Collective rights and the problem of organizing collective action.
 - ▶ Legal certainty.
- ▶ Enough funds, or no financial constraints.
- ▶ Intrinsic costs of negotiation (lawyers, consultants...)
- ▶ Economists refer to all these aspects as **transaction costs**.
- ▶ **Ronald Coase** (1910-2013): property rights can correct externalities if transaction costs are low.



- ▶ The fix is to make agents **internalize** the externality.
- ▶ The private cost then rises towards the social cost (negative externality).
- ▶ We consider four alternatives:
 - ▶ Negotiation (already seen)
(Between private-sector agents).
 - ▶ Regulation of the quantity produced
(Requires government intervention).
 - ▶ Pigouvian tax/Subsidy
(Requires government intervention).
 - ▶ The obligation to compensate the affected party
(Requires government intervention).
- ▶ Each alternative has its advantages and disadvantages, and depends on the context.
- ▶ Could banning the activity that generates the externality be considered as an option?

Government policies: Regulation of production

- ▶ The government can cap the quantity produced or set quotas.
- ▶ What is the ideal quantity of production that the government sets?
 - ▶ The social optimum, where $P = MSC$
- ▶ But in practice it is not easy to **determine**, or to **verify** that the quotas are met.



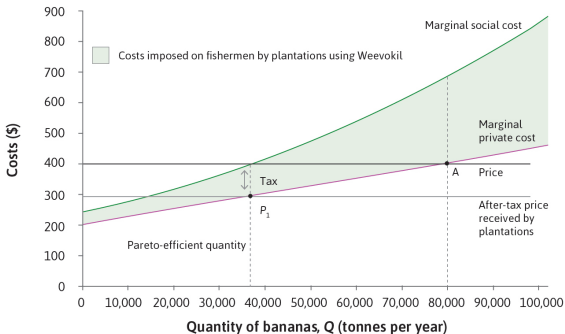
Government policies: The Pigouvian tax

- ▶ Taxes and subsidies also correct externalities:
- ▶ A **Pigouvian tax** equals the (negative) externality: with the tax, marginal private cost equals marginal social cost, so the externality is internalized:
 - ▶ The producer then faces the full cost of the decision.
- ▶ In the case of positive externalities we speak of Pigouvian subsidies.
- ▶ **Arthur Pigou** (1877-1959): English economist, welfare economics, and the tax that bears his name.



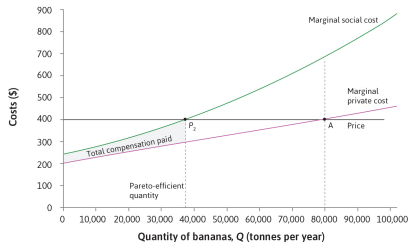
Government policies: The Pigouvian tax

- ▶ What is the amount of the Pigouvian tax?
 - ▶ The Marginal External Cost [MEC].
 - ▶ The farmers then choose the Pareto efficient quantity.



Government policies: Compensation to the affected party

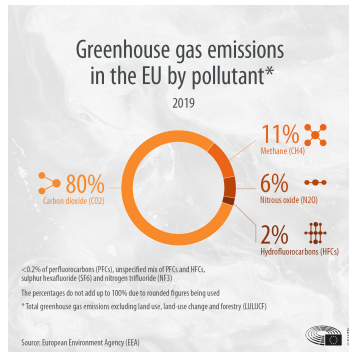
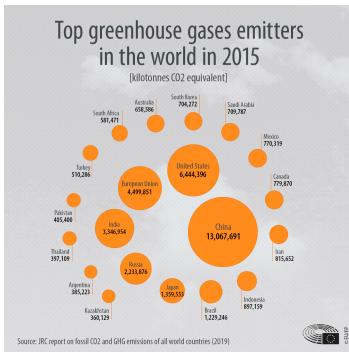
- ▶ If the state makes producers compensate the external cost, the externality is internalized.
- ▶ Compensation increases the marginal cost.
- ▶ If the compensation equals the difference between the marginal social and private cost, then **the quantity produced is efficient**.
- ▶ The effect is similar to the Pigouvian tax if the tax revenue is transferred to those affected.



- ▶ As with negotiation between the parties, government policies to correct the externality have **limitations**:
 - ▶ **Lack of information**: it is difficult to determine the exact compensation to correct the external effects;
 - ▶ **Measurement problems**: the marginal social cost is difficult to calculate;
 - ▶ **Pressure groups**: the government might favor certain groups, creating problems of unfair solutions.

5. Externalities and Climate Change

- Growth since the industrial revolution has come with serious environmental damage.

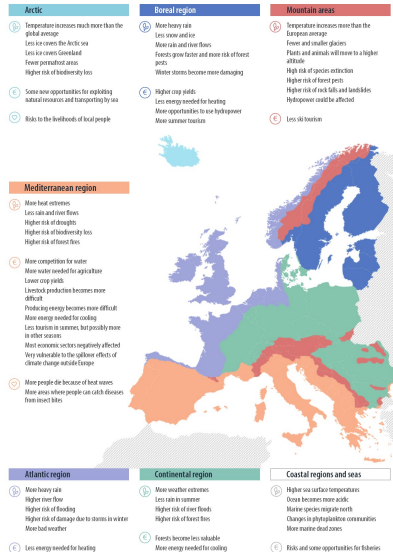


Source: [Climate change Facts and Figures](#), European Parliament.

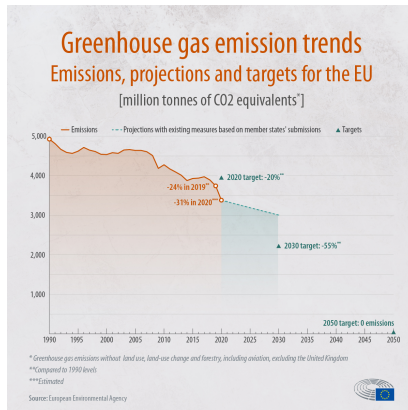
Externalities

The example of climate change

- ▶ Living standards rose, at a heavy external cost.
- ▶ The colors refer to areas, not to the level of risk.

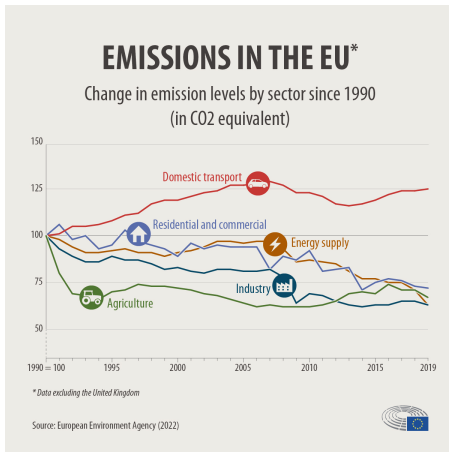


- ▶ What has been done?:
- ▶ EU measures against climate change:
 - ▶ Environmental taxes, tradable permit systems or subsidies targeted at certain groups or activities.
- ▶ In general, in the EU the main policy is based on the “**polluter pays**” principle.



The example of climate change

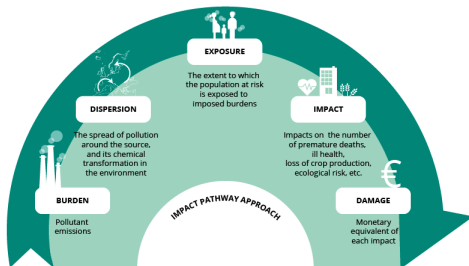
- ▶ But there is still much to be done:



- ▶ The EU has set the goal of being **climate-neutral by 2050**: net zero greenhouse gas emissions.
- ▶ This, as well as the concrete measures to reach the goal, is explained in the **European Green Deal**.

- To mitigate an externality you first have to **measure the external cost**:

Figure 2. Step-by-step schematic of the Impact Pathway Approach framework



Source: Adapted from EEA, 2011; 2014.

- See more details on [this report](#) regarding the measurement of the externalities of industrial pollution.

6. Asymmetric Information

- ▶ With asymmetric information, one party knows something about the transaction that the other does not.
- ▶ There are two forms of **asymmetric information**:
 - ▶ **Hidden actions or moral hazard**: One party carries out actions that the other party does not know about or cannot verify. Example:
 1. The relationship between employer and employee (Micro, Unit 6).
 2. The relationship between voters and elected politicians.
 - ▶ **Hidden attributes or adverse selection**: There are key characteristics known only to one party. Example:
 1. Buyers of second-hand cars do not know all the attributes of the car. The seller has better information;
 2. An insured person with a health policy knows more about their health than the insurance company.
- ▶ If a mutually beneficial exchange does not happen for this reason, we have a **missing market**.

- ▶ In the **private health insurance market** there is **asymmetric information**:
 - ▶ The patient knows their state of health.
 - ▶ The insurance company does not know the client's state of health, there is a **hidden attribute**.
- ▶ The price reflects the insurer's cost, and only those who value the insurance most buy it, the ones in worse health, which pushes cost and price up:
 - ▶ **adverse selection**: only the people in worse health buy the insurance while the healthier ones do not.
- ▶ One solution is for health insurance to be provided by the state (European solution) or provided to groups, e.g. all the workers of a company (North American solution).

- ▶ In the **car insurance market** there is **asymmetric information**:
 - ▶ Insurance companies do not know how the driver behaves (**hidden actions**);
 - ▶ The insured driver has an incentive to be **less careful or prudent**.
- ▶ With fully comprehensive policies the incentive to drive recklessly grows.
- ▶ Solutions:
 - ▶ Insurance companies set limits on the contract and discriminate according to the driver's experience.
 - ▶ Partial insurance (part of the damages is covered by the client)
- ▶ There is a problem of **moral hazard** and of **external effects**.
 - ▶ Careful drivers create positive externalities for the insurance company and for society.

Examples: Banking system

- ▶ In the **credit market** there are many **principal-agent problems** affected by asymmetric information.
- ▶ The borrower's decisions have external effects on the lender.
 - ▶ The client's (borrower's) imprudence limits their ability to meet the loan installments;
 - ▶ To avoid this, banks **restrict access to credit**;
 - ▶ Redistributive effects: people with lower incomes are the most affected.
- ▶ **The banking system as a whole** faces a problem of **moral hazard**.
 - ▶ The problems of one bank spread to others (e.g. through cross-lending), especially if they are large **"too big to fail"**.
 - ▶ The government may be forced to help them in case of problems.
 - ▶ Expecting help, investors and managers take excessive risks: if it goes well they take the profit, if it goes badly the government pays.

"I want to be very, very clear: too big to fail is one of the biggest problems we face in this country, and we must take action to eliminate too big to fail"
(Federal Reserve Chairman 2009, Ben Bernanke)

Imperfect information

Examples: Banking system



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Economy And Business

BANKING BAILOUT +

Spanish bank bailout cost taxpayers €41.8 billion, Audit Court finds

New report says full amount may still climb higher, and confirms that a large chunk is unrecoverable



ÍÑIGO DE BARRÓN

Madrid - JUN 11, 2017 - 10:40 CET



Up until December 2015, the cost of restructuring Spain's bankrupt savings banks had totaled €60.7 billion, of which nearly €41.8 billion was put up by the state through the Orderly Bank Restructuring Fund (FROB) and the rest by the banking sector, according to Spain's Audit Court. In a new report, this public oversight body has found that the real cost of bailing out the sector in the wake of the [Spanish property crash](#) and the global economic crisis will not be fully understood until all the lenders are newly privatized.

- ▶ In the 2008 financial crisis, **many countries bailed out bankrupt banks**;
- ▶ The crisis created unemployment and cut the solvency of many clients.
- ▶ If banks count on public help, they take riskier positions.
- ▶ Banks take advantage of favorable conditions and taxpayers suffer a negative externality.
- ▶ Read the El País newspaper article [here](#).

7. Public Goods

Definition

- ▶ To distinguish **public goods** from **private goods**, and from other types of goods, we use two concepts:
 - ▶ A good is **non-rival** if its use by one person does not reduce its availability to others.
 - ▶ A good is **non-excludable** if it is not possible to exclude any person from enjoying it or accessing it.
- ▶ Combining the four possibilities (rival or not, and excludable or not) we have **four types of goods**:
 - ▶ **Excludable and Rival**: Private goods (*food, clothing, houses, etc.*)
 - ▶ **Excludable and Non-Rival**: artificially scarce **public goods** (*cable TV, uncongested highways, knowledge under intellectual property rights, etc.*)
 - ▶ **Non-Excludable and Rival**: Common resources (*fisheries in a lake, common grazing land for cattle, etc.*)
 - ▶ **Non-Excludable and Non-Rival**: **Public goods** (and public bads) (*watching a lunar eclipse, public television, mathematics, national defense, noise and polluted/clean air*)
- ▶ **Non-rivalry** is the characteristic that defines public goods, which can be **excludable or not**.

- ▶ The market usually allocates **private goods** (Rival and Excludable). If the good is not private, the market fails or does not exist.
- ▶ **Non-rival goods** have a marginal cost equal to zero (the quantity of the good is not reduced by its consumption):
 - ▶ In a competitive market $P = MC$ makes the price zero, so nobody produces them: non-rival goods have no competitive market (unless subsidised).
 - ▶ Firms with market power can provide them and make a profit, but market power creates other problems.
- ▶ **Non-excludable goods** cannot be given a price:
 - ▶ Price is the way to exclude those who could consume the good (without paying you cannot access the consumption of the good).

8. Limits to Markets

- ▶ The market is the usual **mechanism to allocate goods**: **unlimited needs, scarce resources**.
- ▶ The great advantage of the market is that it achieves a **Pareto efficient allocation**.
- ▶ But it has **limitations**:
 - ▶ Unregulated markets can lead to allocations that are not **Pareto efficient** (market power, externalities, etc.)
 - ▶ Other ways of carrying out the allocation (such as through the government or families) may be more effective;
- ▶ The market mechanism may leave out **social preference norms**:
 - ▶ For example, paying students to obtain good grades.
 - ▶ **Repugnant markets**: markets that violate ethical or good social conduct norms, such as slavery.
 - ▶ **Merit goods**: they should be provided to everyone, regardless of their economic means (education, health, etc.)

► Watch this [video](#).

Should we trust markets?



9. Summary

Today's class, section by section:

1. **Introduction** → a market failure is an allocation that is not the best possible one.
2. **Externalities: definition** → effects on others, in production or consumption, that no contract covers.
3. **Externalities: analysis** → $MSC = MPC + MEC$; a negative externality means $MSC > MPC$ and overproduction.
4. **Externalities: solutions** → negotiation, regulation, Pigouvian tax or compensation, all of them to internalize the cost.
5. **Climate change** → the externality of the century: the “polluter pays” principle and the problem of measuring the cost.
6. **Asymmetric information** → hidden actions (moral hazard) and hidden attributes (adverse selection) create missing markets.
7. **Public goods** → non-rival goods have zero marginal cost, so no competitive market provides them.
8. **Limits to markets** → efficiency is not everything: social norms, repugnant markets and merit goods.

Key concepts: market failure, externality, MPC, MEC, MSC, transaction costs, Coase, Pigouvian tax, moral hazard, adverse selection, missing market, non-rival, non-excludable, merit good.